

Problem 8.3

The amount of interest earned on A for one year is \$562.50, while the equivalent amount of discount is \$500. Find A .

Solution

Let i be the effective rate of interest and d the effective rate of discount for one year.

Since the amount of interest earned on A for one year is \$562.50,

$$Ai = 562.50,$$

so

$$i = \frac{562.50}{A}.$$

Since the equivalent amount of discount is \$500,

$$Ad = 500,$$

so

$$d = \frac{500}{A}.$$

Using the relationship between i and d ,

$$i = \frac{d}{1 - d}.$$

Substitute the expressions for i and d :

$$\frac{562.50}{A} = \frac{\frac{500}{A}}{1 - \frac{500}{A}}.$$

Multiply both sides by A :

$$562.50 = \frac{500}{1 - \frac{500}{A}}.$$

Now invert the equation:

$$\frac{500}{562.50} = 1 - \frac{500}{A}.$$

Since

$$\frac{500}{562.50} = \frac{8}{9},$$

we get

$$\frac{8}{9} = 1 - \frac{500}{A}.$$

Thus,

$$\frac{500}{A} = 1 - \frac{8}{9} = \frac{1}{9}.$$

So,

$$A = 500 \cdot 9 = 4500.$$

Therefore,

$$\boxed{A = 4500}.$$